

CSE213

MICROCONTROLLER PROGRAMMING

Memory Interface

Introduction

- Simple or complex, every microprocessor-based system has a memory system.
- Almost all systems contain two main types of memory: **read-only memory** (ROM) and **random access memory** (RAM) or read/write memory.
- This chapter explains how to interface both memory types to the Intel family of microprocessors.

Chapter Objectives

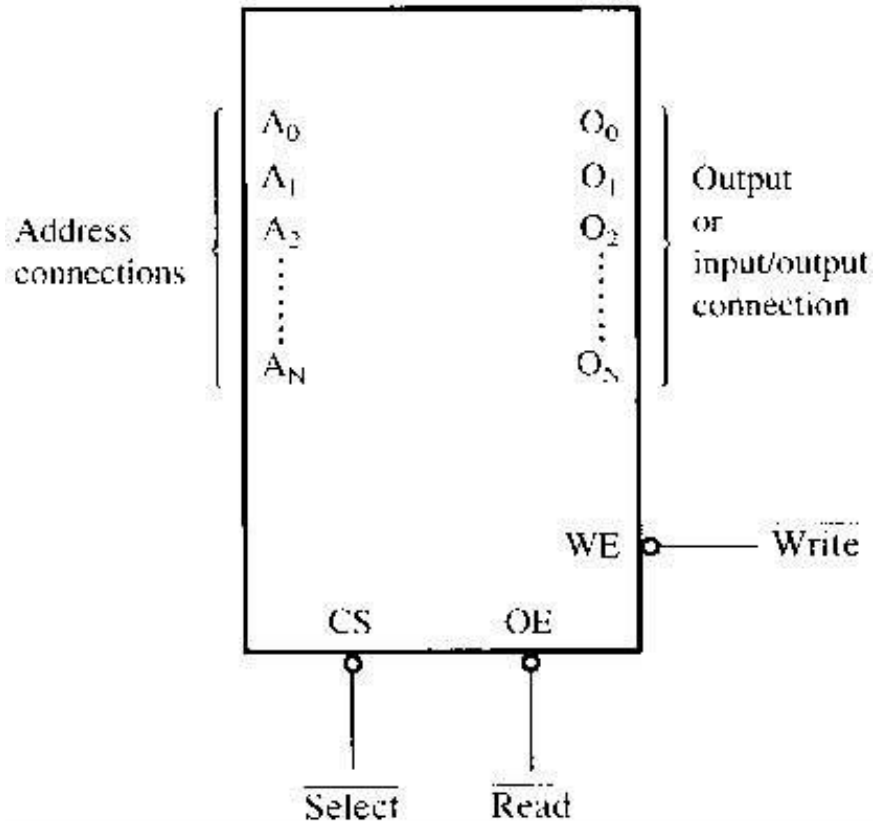
Upon completion of this chapter, you will be able to:

- Decode the memory address and use the outputs of the decoder to select various memory components.
- Use programmable logic devices (PLDs) to decode memory addresses.
- Explain how error correction code (ECC) is used with memory.

MEMORY DEVICES

- Before attempting to interface memory to the microprocessor, it is essential to understand the operation of memory components.
- In this section, we explain functions of the four common types of memory:
 - Read-Only Memory (ROM)
 - Flash Memory (EEPROM)
 - Static Random Access Memory (SRAM)
 - Dynamic Random Access Memory (DRAM)

Memory Pin Connections



- address inputs
- data outputs or input/outputs
- some type of selection input
- at least one control input to select a read or write operation

Figure. A pseudomemory component illustrating the address, data, and control connections.

Address Connections

- Memory devices have address inputs to select a memory location within the device.
- Almost always labeled from A_0 , the least significant address input, to A_n
 - where subscript n can be any value
 - always labeled as one less than total number of address pins
- A memory device with 10 address pins has its address pins labeled from A_0 to A_9 .

- The number of address pins on a memory device is determined by the number of memory locations found within it.
- Today, memory devices have memory locations starting from 1K to multiple G.
- A 1K memory device has 10 address pins.
 - therefore, 10 address inputs are required to select any of its 1024 memory locations

- It takes a 10-bit binary number to select any single location on a 1024-location device.
 - 1024 different combinations
 - if a device has 11 address connections, it has 2048 (2K) internal memory locations
- The number of memory locations can be extrapolated from the number of pins.

Data Connections

- All memory devices have a set of data outputs or input/outputs.
 - today, many devices have bidirectional common I/O pins
 - data connections are points at which data are entered for storage or extracted for reading
- Data pins on memory devices are labeled D_0 through D_7 for an 8-bit-wide memory device.

- An 8-bit-wide memory device is often called a **byte-wide** memory.
 - most devices are currently 8 bits wide,
 - some are 16 bits, 4 bits, or just 1 bit wide
- Catalog listings of memory devices often refer to memory locations times bits per location.
 - a memory device with 1K memory locations and 8 bits in each location is often listed as a **1K × 8** by the manufacturer
- Memory devices are often classified according to total bit capacity.

Selection Connections

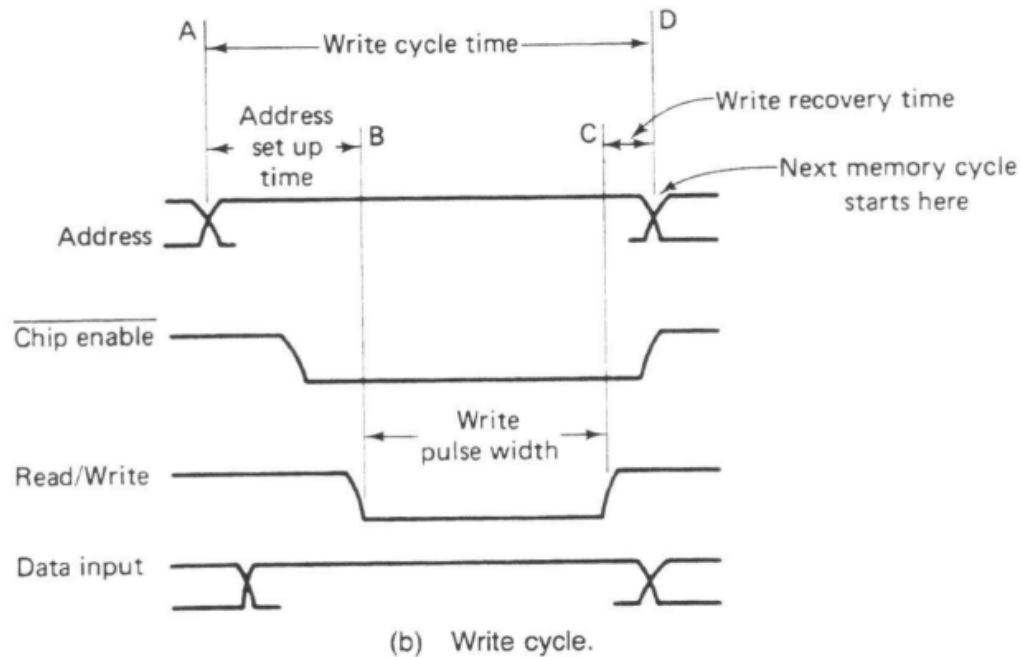
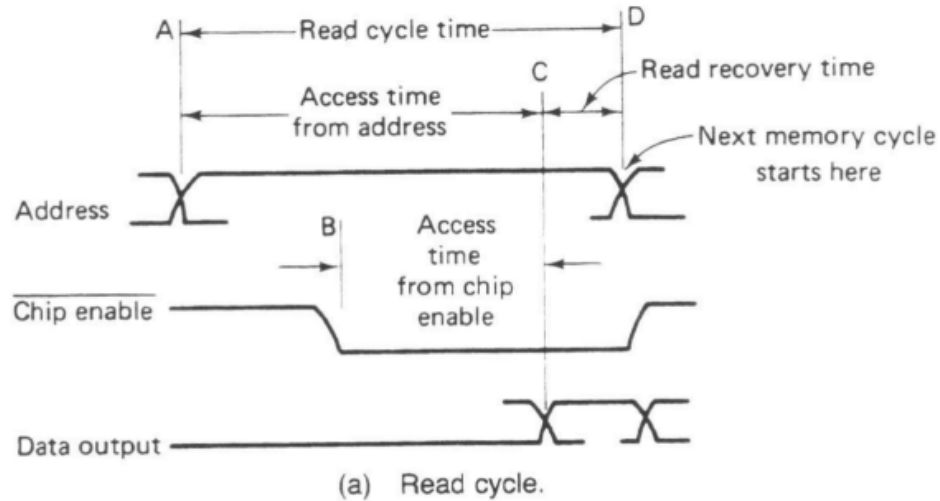
- Each memory device has an input that selects or enables the memory device.
 - sometimes more than one
- This type of input is most often called a **chip select** ($\overline{CS}$) **chip enable** ($\overline{CE}$) or simply **select** ($\overline{S}$) input.
- RAM memory generally has at least one $\overline{CS}$ or $\overline{S}$ input, and ROM has at least one $\overline{CE}$
- If more than one $\overline{CE}$ connection is present, all must be activated to read or write data.

Control Connections

- All memory devices have some form of control input or inputs.
 - ROM usually has one control input, while RAM often has one or two control inputs
- Control input often found on ROM is the **output enable** or **gate** connection, which allows data flow from output data pins.
- The $\overline{\text{OE}}$ connection enables and disables a set of three-state buffers located in the device and must be active to read data.

- RAM has either one or two control inputs.
 - if one control input, it is often called $R/\overline{W}$
- If the RAM has two control inputs, they are usually labeled $\overline{WE}$ (or $\overline{W}$), and $\overline{OE}$ (or $\overline{G}$).
 - **write enable** must be active to perform memory write, and $\overline{OE}$ active to perform a memory read
 - when the two controls are present, they must never both be active at the same time
- If both inputs are inactive, data are neither written nor read.
 - the connections are at their high-impedance state

Sample Memory Read/Write Timing Cycles



Datasheets

- You can get detailed information on ICs including memory units by just looking at their datasheets.
- Every manufacturer releases datasheets of their products which includes operational information.

ROM Memory

- **Read-only memory (ROM)** permanently stores programs/data resident to the system.
 - and must not change when power disconnected
- Often called **nonvolatile memory**, because its contents *do not* change even if power is disconnected.
- A device we call a ROM is purchased in mass quantities from a manufacturer.
 - programmed during fabrication at the factory

- The EPROM (**erasable programmable read-only memory**) is commonly used when software must be changed often.
 - or when low demand makes ROM uneconomical
 - for ROM to be practical at least 10,000 devices must be sold to recoup factory charges
- An EPROM is programmed in the field on a device called an EPROM programmer.
- Also erasable if exposed to high-intensity ultraviolet light.
 - depending on the type of EPROM

- PROM memory devices are also available, although they are not as common today.
- The PROM (**programmable read-only memory**) is also programmed in the field by burning open tiny Ni-chrome or silicon oxide fuses.
- Once it is programmed, it cannot be erased.

- A newer type of **read-mostly memory** (RMM) is called the **flash memory**.
 - also often called an EEPROM (**electrically erasable programmable ROM**)
 - EAROM (**electrically alterable ROM**)
 - or a NOVRAM (**nonvolatile RAM**)
- Electrically erasable in the system, but they require more time to erase than normal RAM.
- The flash memory device is used to store setup information for systems such as the video card in the computer.

- Flash has all but replaced the EPROM in most computer systems for the BIOS.
 - some systems contain a password stored in the flash memory device
- Flash memory has its biggest impact in memory cards for digital cameras and memory in MP3 audio players.

Static RAM (SRAM) Devices

- Static RAM memory devices retain data as long as DC power is applied.
- Because no special action is required to retain data, these devices are called **static memory**.
 - also called **volatile memory** because they will not retain data without power
- The main difference between ROM and RAM is that RAM is written under normal operation, whereas ROM is programmed outside the computer and normally is only read.

- Access time on a typical 4016 SRAM is 250 ns, fast enough to connect directly to an 8088/8086 at 5 MHz, without wait states.
- Access time must always be checked to determine compatibility of memory components
- Access times can be as low as 1.0 ns for SRAM used in computer cache memory

Dynamic RAM (DRAM) Memory

- Available up to $256\text{M} \times 8$ (2G bits).
- DRAM is essentially the same as SRAM, except that it retains data for only 2 or 4 ms on an integrated capacitor.
- After 2 or 4 ms, the contents of the DRAM must be completely rewritten (*refreshed*).
 - because the capacitors, which store a logic 1 or logic 0, lose their charges

- In DRAM, the entire contents are refreshed with 256 reads in a 2- or 4-ms interval.
 - also occurs during a write, a read, or during a special refresh cycle
- DRAM requires so many address pins that manufacturers multiplexed address inputs.
- DRAM is often placed on small boards called SIMMs (Single In-Line Memory Modules) or DIMMs (Dual In-Line Memory Modules).

- Another type is the RIMM memory module from RAMBUS Corporation,
 - this memory type has faded from the market
- The latest DRAM is the DDR (**double-data rate**) memory device and DDR2.
 - DDR transfers data at each edge of the clock, making it operate at twice the speed of SDRAM.



Common DRAM packages

From top to bottom:
DIP, SIMM, SIMM
(30-pin), SIMM (72-
pin), DIMM (168-
pin), DDR DIMM
(184-pin).

SRAM vs DRAM

- SRAM is static while DRAM is dynamic
- SRAM is faster compared to DRAM
- SRAM consumes less power than DRAM
- SRAM uses more transistors per bit of memory compared to DRAM
- SRAM is more expensive than DRAM
- Cheaper DRAM is used in main memory while SRAM is commonly used in cache memory

ADDRESS DECODING

- In order to attach a memory device to the microprocessor, it is necessary to decode the address sent from the microprocessor.
- Decoding makes the memory function at a unique section or partition of the memory map.
- Without an address decoder, only one memory device can be connected to a microprocessor, which would make it virtually useless.

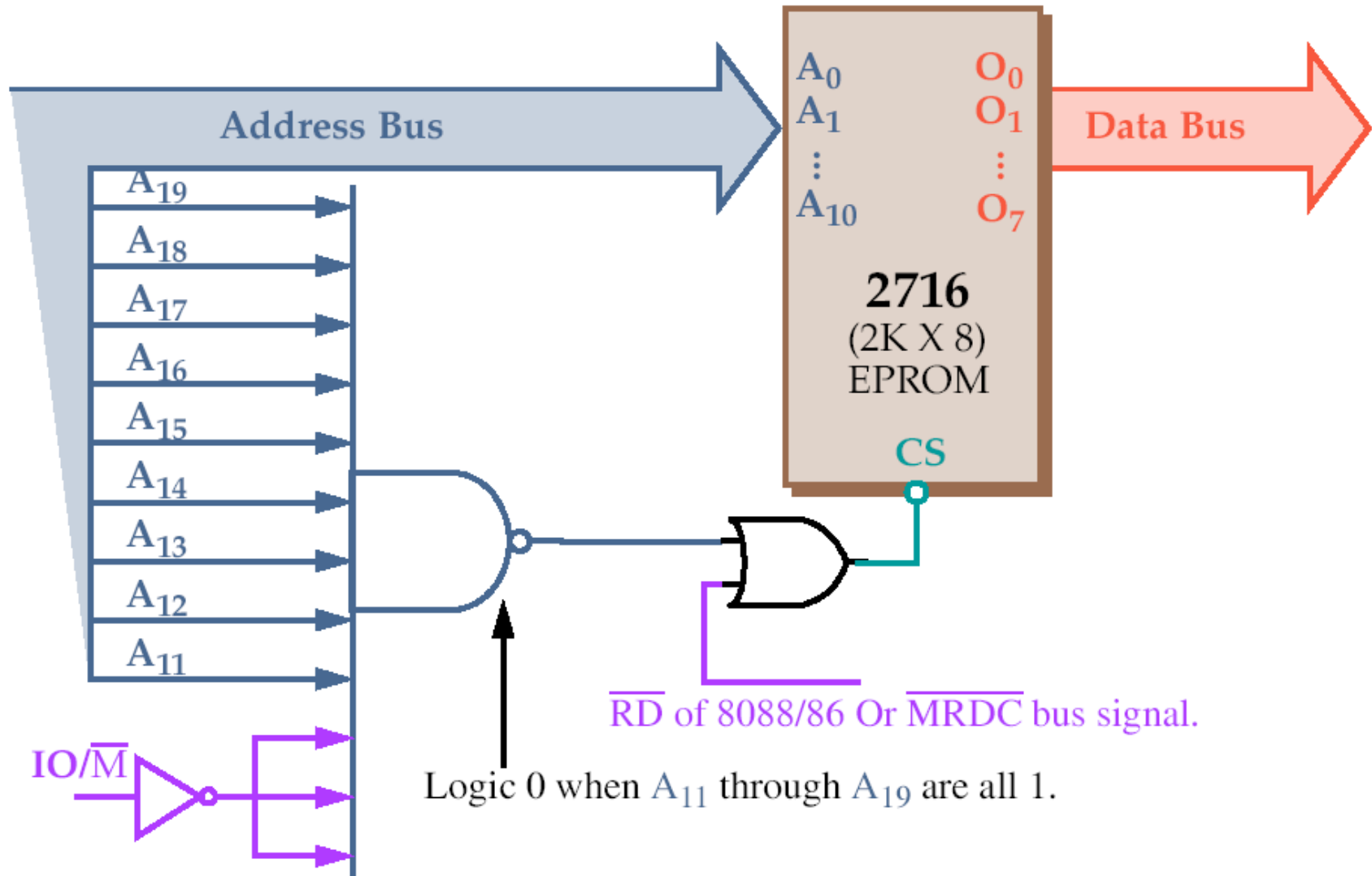
Why Decode Memory?

- The 8088 has 20 address connections and the 2716 EPROM has 11 connections.
- The 8088 sends out a 20-bit memory address whenever it reads or writes data.
 - because the 2716 has only 11 address pins, there is a mismatch that must be corrected
- The decoder corrects the mismatch by decoding address pins that do not connect to the memory component.

Simple NAND Gate Decoder

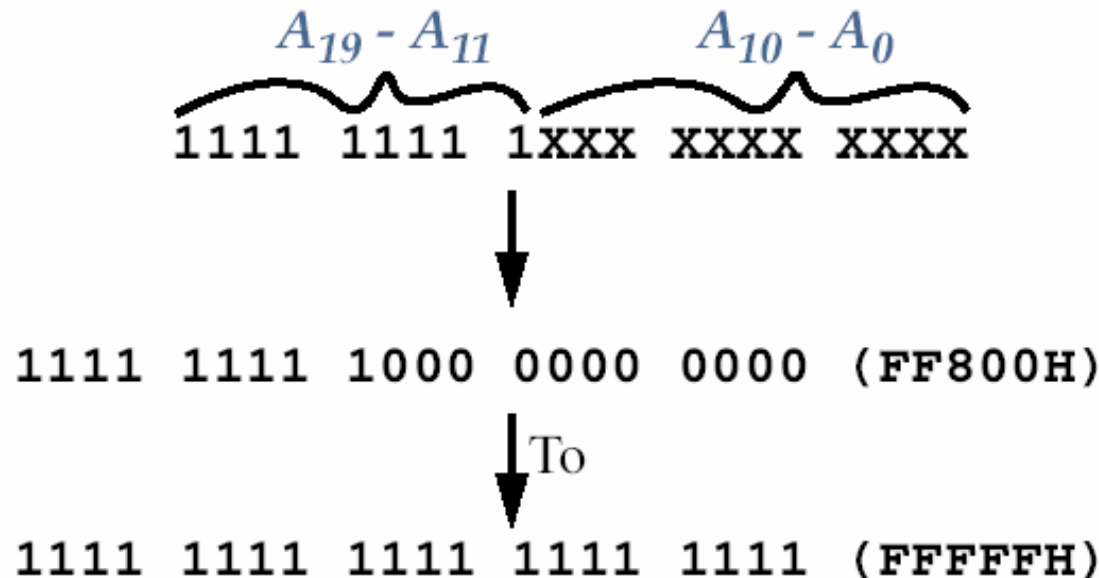
- When the $2K \times 8$ EPROM is used, address connections $A_{10}-A_0$ of 8088 are connected to address inputs $A_{10}-A_0$ of the EPROM.
 - the remaining nine address pins ($A_{19}-A_{11}$) are connected to a NAND gate decoder
- The decoder selects the EPROM from one of the 2K-byte sections of the 1M-byte memory system in the 8088 microprocessor.
- In this circuit a NAND gate decodes the memory address, as seen in the figure.

Figure. A simple NAND gate decoder that selects a 2716 EPROM for memory location FF800H–FFFFFH.



- If the 20-bit binary address, decoded by the NAND gate, is written so that the leftmost nine bits are 1s and the rightmost 11 bits are don't cares (X), the actual address range of the EPROM can be determined.
 - a *don't care* is a logic 1 or a logic 0, whichever is appropriate

To determine the address range that a device is mapped into:



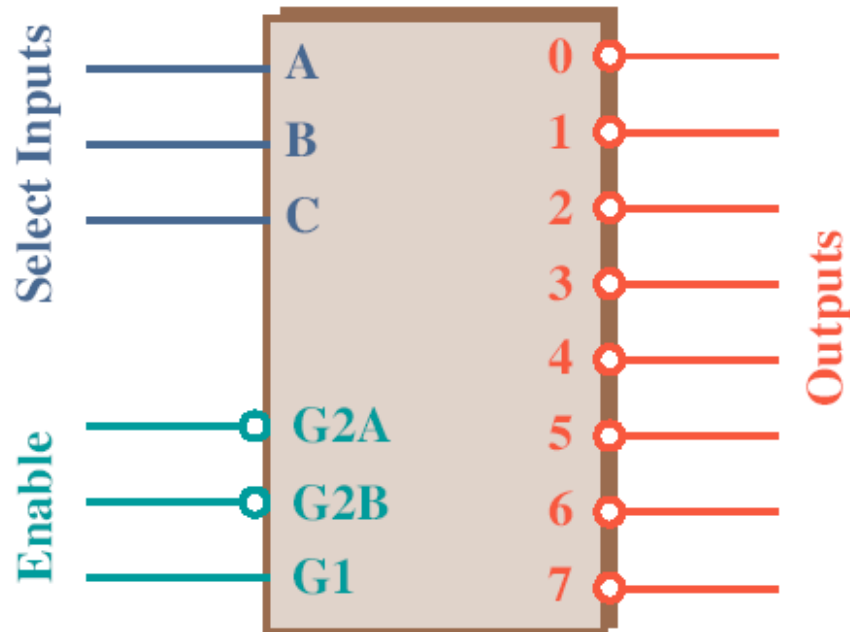
NAND gate decoders are not often used

Large fan-in NAND gates are not efficient

Multiple NAND gate IC's might be required to perform such decoding

Rather the 3-to-8 Line Decoder (74LS138) is more common.

The 3-to-8 Line Decoder (74LS138)

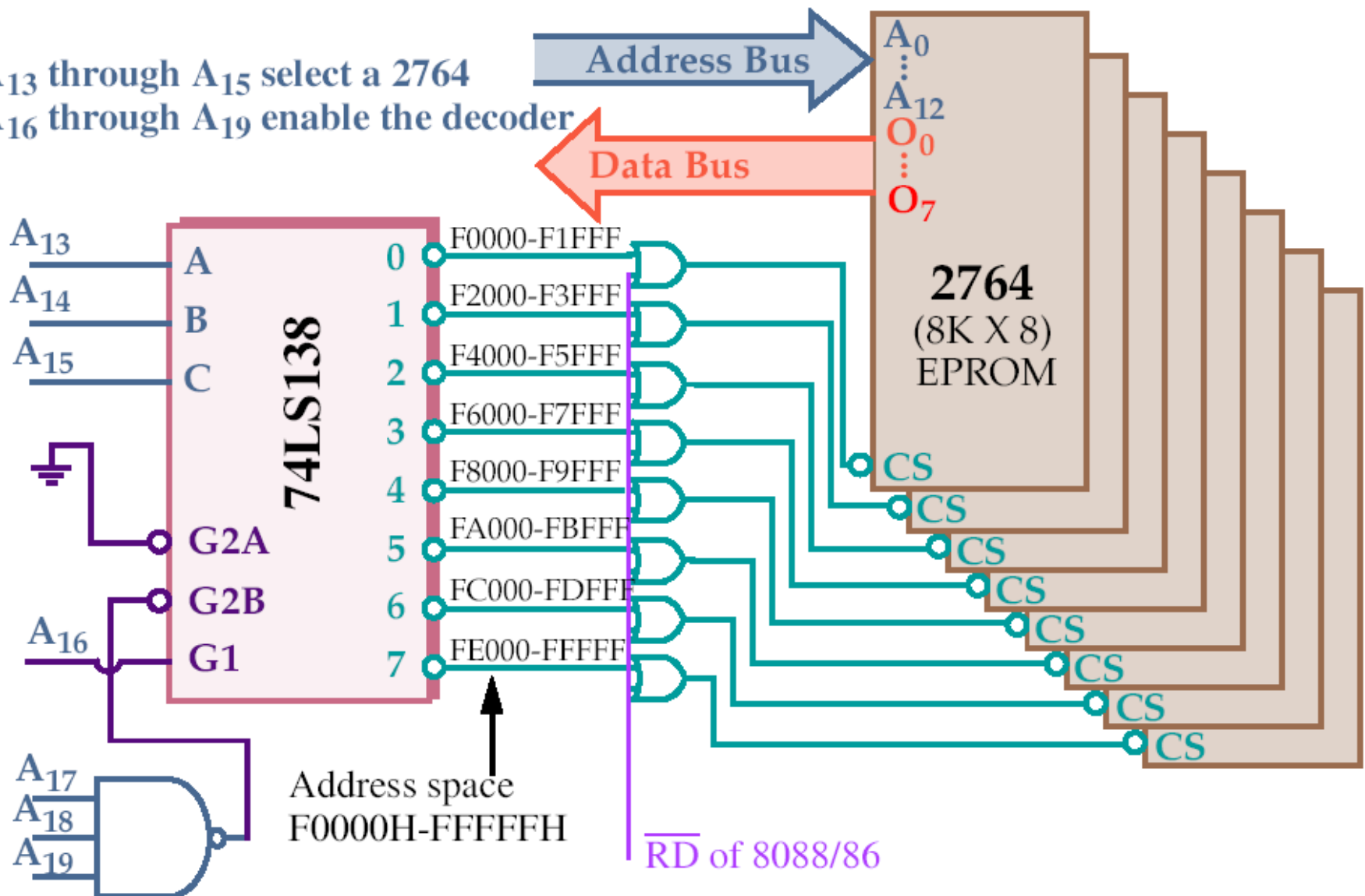


Inputs						Output							
Enable			Select										
G2A	G2B	G1	C	B	A	0	1	2	3	4	5	6	7
1	X	X	X	X	X	1	1	1	1	1	1	1	1
X	1	X	X	X	X	1	1	1	1	1	1	1	1
X	X	0	X	X	X	1	1	1	1	1	1	1	1
0	0	1	0	0	0	0	1	1	1	1	1	1	1
0	0	1	0	0	1	1	0	1	1	1	1	1	1
0	0	1	0	1	0	1	1	0	1	1	1	1	1
0	0	1	0	1	1	1	1	1	0	1	1	1	1
0	0	1	1	0	0	1	1	1	1	0	1	1	1
0	0	1	1	0	1	1	1	1	1	1	0	1	1
0	0	1	1	1	0	1	1	1	1	1	1	0	1
0	0	1	1	1	1	1	1	1	1	1	1	1	0

Note that all *three* Enables (G2A, G2B, and G1) must be active, e.g. low, low and high, respectively.

Each output of the decoder can be attached to an 2764 EPROM (8K X 8).

A_{13} through A_{15} select a 2764
 A_{16} through A_{19} enable the decoder



The EPROMs cover a 64KB section of memory.

PLD Programmable Decoders

- Three SPLD (**simple PLD**) devices function in the same manner but have different names:
 - PLA (**programmable logic array**)
 - PAL (**programmable array logic**)
 - GAL (**gated array logic**)
- In existence since the mid-70s, they have appeared in memory system and digital designs since the early 1990s.

- PAL and PLA are fuse-programmed, and some PLD devices are erasable devices.
 - all are arrays of programmable logic elements
- Other PLDs available:
 - CPLDs (**complex programmable logic devices**)
 - FPGAs (**field programmable gate arrays**)
 - FPICs (**field programmable interconnect**)
- These PLDs are more complex than the SPLDs used more commonly in designing a complete system.

- If the concentration is on decoding addresses, the SPLD is used.
- If the concentration is on a complete system, then the CPLD, FPGA, or FPIC is used to implement the design.
- These devices are also referred to as an ASIC (**application-specific integrated circuit**).

Error Detection and Correction

- ***Parity, BCC (Block-Check Character) and CRC (Cyclic Redundancy Check)*** are only mechanisms for error detection.
- The system is halted if an error is found in memory.
- Error *correction* is starting to show up in new systems.
- ***SDRAM*** has ***ECC (Error Correction Code)***.
- Correction will allow the system can continue operating.
- If *two* errors occur, they can be *detected* but not *corrected*.
- Error correction will of course cost more in terms of extra bits.

Error Correction

- Error correction is based on Hamming codes
- Error detection and correction
- Presumes one bit error most likely
- Detects error and position
therefore allows for correction

Hamming Code Example

Data is on bits 3, 5, 6, 7, 9, 10, 11, 12

Bit position

1	2	3	4	5	6	7	8	9	10	11	12
P_1	P_2	1	P_4	1	0	0	P_8	0	1	0	0

$$P_1 = \text{XOR of bits (3, 5, 7, 9, 11)} = 1 \oplus 1 \oplus 0 \oplus 0 \oplus 0 = 0$$

$$P_2 = \text{XOR of bits (3, 6, 7, 10, 11)} = 1 \oplus 0 \oplus 0 \oplus 1 \oplus 0 = 0$$

$$P_4 = \text{XOR of bits (5, 6, 7, 12)} = 1 \oplus 0 \oplus 0 \oplus 0 = 1$$

$$P_8 = \text{XOR of bits (9, 10, 11, 12)} = 0 \oplus 1 \oplus 0 \oplus 0 = 1$$

Bit position

1	2	3	4	5	6	7	8	9	10	11	12
0	0	1	1	1	0	0	1	0	1	0	0

This is what is transmitted or stored

Note that parity bits are in power of 2 positions

Hamming Code Example

If data received or read from memory is correct

Bit position	1	2	3	4	5	6	7	8	9	10	11	12
	0	0	1	1	1	0	0	1	0	1	0	0

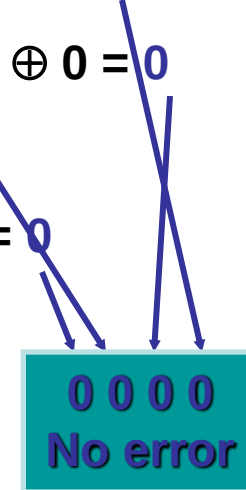
4 check bits are evaluated as follows:

$$C_1 = \text{XOR of bits (1, 3, 5, 7, 9, 11)} = 0 \oplus 1 \oplus 1 \oplus 0 \oplus 0 \oplus 0 = 0$$

$$C_2 = \text{XOR of bits (2, 3, 6, 7, 10, 11)} = 0 \oplus 1 \oplus 0 \oplus 0 \oplus 1 \oplus 0 = 0$$

$$C_4 = \text{XOR of bits (4, 5, 6, 7, 12)} = 1 \oplus 1 \oplus 0 \oplus 0 \oplus 0 = 0$$

$$C_8 = \text{XOR of bits (8, 9, 10, 11, 12)} = 1 \oplus 0 \oplus 1 \oplus 0 \oplus 0 = 0$$



0000
No error

Hamming Code Example

If data received or read from memory is not correct

Bit position	1	2	3	4	5	6	7	8	9	10	11	12
	0	0	1	1	1	0	0	1	0	1	0	0

If read incorrectly (change bit 3 from a 1 to 0)

Bit position	1	2	3	4	5	6	7	8	9	10	11	12
	0	0	0	1	1	0	0	1	0	1	0	0

4 check bits are evaluated as follows:

$$C_1 = \text{XOR of bits (1, 3, 5, 7, 9, 11)} = 0 \oplus 0 \oplus 1 \oplus 0 \oplus 0 \oplus 0 = 1$$

$$C_2 = \text{XOR of bits (2, 3, 6, 7, 10, 11)} = 0 \oplus 0 \oplus 0 \oplus 0 \oplus 1 \oplus 0 = 1$$

$$C_4 = \text{XOR of bits (4, 5, 6, 7, 12)} = 1 \oplus 1 \oplus 0 \oplus 0 \oplus 0 = 0$$

$$C_8 = \text{XOR of bits (8, 9, 10, 11, 12)} = 1 \oplus 0 \oplus 1 \oplus 0 \oplus 0 = 0$$

0 0 1 1
Error
in bit 3



The Intel Microprocessors: 8086/8088, 80186/80188, 80286, 80386, 80486 Pentium, Pentium Pro Processor, Pentium II, Pentium, 4, and Core2 with 64-bit Extensions Architecture, Programming, and Interfacing, Eighth Edition
Barry B. Brey

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